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Spatiotemporal assessment of urban environmental livability in Dhaka (2004–2024)

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Rapid and largely unplanned urbanization has significantly reshaped Dhaka over the past two decades. This transformation has accelerated environmental degradation, reduced green and blue spaces, and undermined urban livability in one of the world's most densely populated megacities. Despite these growing environmental challenges, comprehensive long-term spatial assessments that integrate remote sensing and multi-criteria decision-making approaches remain scarce. This study assesses the spatiotemporal dynamics of urban environmental livability in Dhaka for 2004, 2014, and 2024 using the Urban Environmental Livability Index (UELI). Six indicators, including the Normalized Difference Vegetation Index (NDVI), Wetness, Index-based Built-up Index (IBI), Night Light Index (NLI), Urban Heat Island Ecological Intensity Index (UHIEII), and Air Pollution Index (API), were derived from multi-sensor remote sensing datasets in Google Earth Engine (GEE). Principal Component Analysis (PCA) and the Fuzzy Analytic Hierarchy Process (FAHP) were employed to determine indicator weights, while Global Moran's I and Local Indicators of Spatial Association (LISA) were used to analyze the spatial distribution and clustering of UELI. The findings reveal a substantial decline in environmental livability, with livable areas decreasing from 30.53% in 2004 to 26.87% in 2014 and further to 20.75% in 2024. In contrast, unlivable areas expanded from 15.65% to 17.84% and 30.80% over the same period. Notably, approximately 24.69% of Dhaka's total area transitioned to an unlivable state between 2004 and 2024. One-at-a-Time (OAT) sensitivity analysis and Partial Rank Correlation Coefficient (PRCC) analysis with a 95% confidence interval confirmed the robustness of the UELI framework and highlighted the dynamic nature of urban-environment interactions. Spatial analysis further identified the central and southwestern parts of Dhaka as the most affected areas, exhibiting the greatest deterioration in environmental livability and ecological conditions. The study provides robust empirical evidence to support sustainable urban planning, informed policy formulation, and effective urban ecological risk mitigation in rapidly urbanizing cities.

Keywords Urban environmental livability index (UELI), Multi-source remote sensing, Fuzzy analytic hierarchy process (FAHP), Spatial autocorrelation, Sensitivity analysis, Dhaka megacity

Urbanization has become one of the most transformative global processes, reshaping population distribution, settlement expansion, infrastructure development, and economic growth. Since 1950, the world has experienced unprecedented growth in urban population concentration[1]. While urbanization contributes significantly to human and economic development[2], its rapid and uncontrolled expansion often disrupts urban ecological systems through intensified human activities and land-use change[3–5].

Urban livability is widely recognized as an indicator of the quality of the human environment, encompassing physical, cultural, social, and environmental dimensions that influence human well-being[6–8]. Sustainable urban livability extends this concept by promoting improvements in key urban dimensions (e.g., physical, economic, and socio-cultural) without compromising environmental integrity or the quality of life of residents[9]. As a core principle of sustainable urbanism, the integration of environmental and socio-economic dimensions seeks to balance ecological sustainability with socio-economic benefits[10]. Consequently, sustainable urban development requires continuous evaluation of urban ecological conditions and their evolution under changing urban planning and settlement patterns[11].

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Developing countries increasingly recognize livability assessment as a critical component of sustainable urban development[7,12,13]. As one of the fastest urbanizing developing countries in South Asia, Bangladesh has experienced substantial rural-to-urban migration driven by expanding economic opportunities and improved access to urban services[14–16]. Across South Asian megacities, several regional studies have systematically evaluated urban environmental livability, microclimatic variance, and sustainability at various spatial scales using diverse indicators, such as the Urban Livability Index (ULI)[17], Metropolitan Urban Agglomeration (MUA) frameworks[18], the Sustainable ULI model through Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA) approaches[19], and the Urban Livability Composite Index (ULCI)[20], Urban Thermal Field Variance Index (UTFVI)[21]. Collectively, these studies demonstrate the growing importance of integrated indicator-based approaches for evaluating urban sustainability and environmental livability. However, substantial differences remain in indicator selection, weighting strategies, spatial scales, and validation procedures, limiting the comparability and transferability of findings across rapidly urbanizing cities.

Among major South Asian urban centers, Dhaka's livability is under mounting pressure from rapid, unplanned urbanization, worsening environmental degradation, poor air quality, and an intensifying urban heat-island effect that together compound the city's ecological risk[22]. With over 24 million residents as of 2024, Dhaka has grown from a population of roughly 3 million in the 1980s into one of the world's most densely populated cities[23] and this concentration of people, economic activity, and informal settlements sets it apart from other major Bangladeshi cities such as Khulna, Rajshahi, and Chattogram, making its livability and sustainability challenges especially acute[24]. While a substantial body of research has examined individual dimensions of this crisis, such as air pollution[25], flooding[26], urban heat island dynamics[27], traffic congestion[28], and climate vulnerability[29,30] these threads are rarely woven into a single, spatially explicit framework that links environmental quality directly to urban livability. Furthermore, most existing studies in Dhaka focus on individual environmental issues rather than providing a comprehensive assessment that integrates environmental quality indicators, spatial clustering analysis, and scientifically validated weighting approaches within a unified urban environmental livability framework.

To bridge this gap, modern satellite remote sensing technologies offer an efficient way to evaluate urban eco-livability using data on land use, vegetation cover, and surface temperature. Even though conventional techniques are still crucial, assessments based on remote sensing offer significant assistance for well-informed government decision-making[7]. Air Quality Index (AQI), land surface temperature (LST), and Night Time Light (NTL) signify vital parameters to indicate levels of urban livability by evaluating air pollution, thermal conditions, and built environments[31–33]. Many Scholars focus on urban livability assessing the diverse urban environmental aspects to develop multidimensional integrated indices[34]. The indices such as, the Urban Environmental Quality Index (UEQI)[35], Urban Ecological Livability Index (UELI)[11], Livable Level Integrated Index (LLII)[36], and the Urban Environment and Social Inclusion Index (UESI)[37]. These frameworks have substantially improved the assessment of urban environmental conditions; however, their applicability often depends on data availability, indicator selection, and methodological assumptions.

However, despite the advantages of cloud computing platforms such as GEE, existing remote sensing-based urban environmental assessments face several challenges, including dependency on single-source data, required human processing, insufficient scientific validation, and limited integration of multi-source datasets[7]. Constructing an objective and extensive UELI using remote sensing data is crucial to more successfully addressing the aforementioned problems. It is important to emphasize that this study specifically calculates the UELI to evaluate the environmental and biophysical dimensions of urban quality, rather than a broad ULI which would require non-environmental components such as social governance, healthcare, and municipal infrastructure.

To address these empirical and methodological limitations, this study proposes an integrated urban environmental livability framework for Dhaka, which faces significant impacts of rapid and unplanned urbanization that deteriorate the sustainability of both natural and built-up environments[38]. In this study, all UELI indicators were derived and processed using the GEE platform and integrated through a hybrid PCA–FAHP framework. The resulting index was further evaluated using sensitivity analysis and spatial autocorrelation techniques to ensure robustness and spatial reliability. Given Dhaka's rapid urban expansion and growing environmental challenges, a scientifically robust assessment of urban environmental livability is essential for supporting evidence-based urban planning and sustainable development policies. Such assessments can assist policymakers in identifying environmentally vulnerable areas, prioritizing resource allocation, and mitigating long-term ecological risks.

Therefore, the primary objective of this study is to assess the spatiotemporal dynamics of urban environmental livability in Dhaka between 2004 and 2024 through the development of a comprehensive UELI. By integrating multi-source remote sensing datasets, hybrid weighting techniques, sensitivity analysis, and spatial autocorrelation approaches, the study aims to provide a scientifically validated framework for evaluating environmental livability and supporting sustainable urban planning in rapidly urbanizing megacities.

Methodology

Study area

This study focuses on Dhaka, the capital city of Bangladesh. Geographically located between latitudes 23.58°N and 23.90°N, and longitudes 90.33°E and 90.50°E (Fig. 1)[39]. Dhaka, located in the central part of the Ganges–Meghna–Brahmaputra delta plain, is surrounded by four major rivers: the Buriganga, Turag, Tongi, and Balu[40].

The Dhaka city covers a total area of approximately 306.4 square kilometers with a large number of population which is 22.4 million[41,42]. The present average annual population growth rate in urban area of Dhaka is 1.67% compared with the national annual average 3.90%[41]. The climatic conditions of Dhaka is humid under the sub-tropical monsoon zone and annually receives more than 80% precipitation which is 2000 mm occurs during the

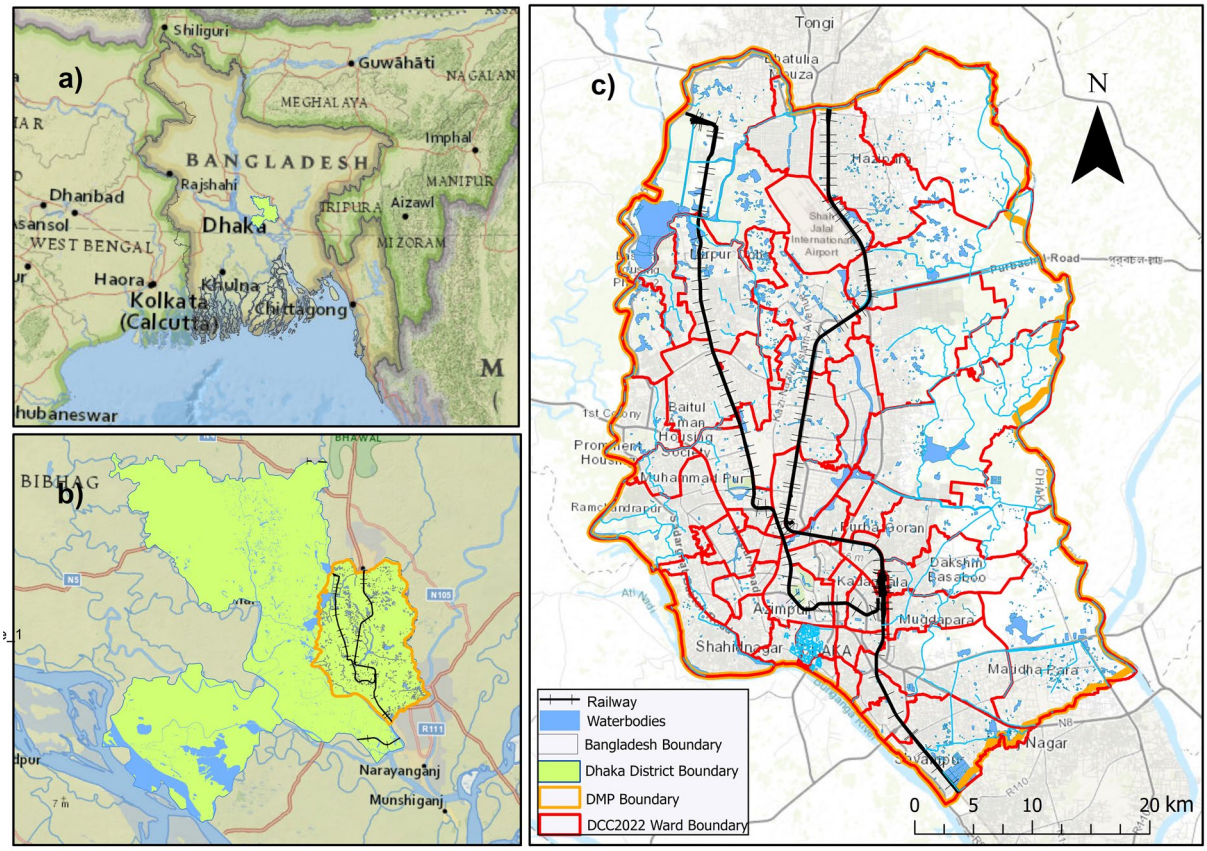


Fig. 1. Geographical location of (**a**) Bangladesh, (**b**) Dhaka District, and (**c**) Dhaka City Corporation. The figure was prepared by the authors using ArcGIS Desktop 10.8 (ArcMap), Esri (https://www.esri.com/arcgis), with an Esri ArcGIS Online basemap (https://www.arcgis.com/) and administrative boundary data from the Bangladesh Bureau of Statistics (BBS). Railway and waterbody layers were obtained from OpenStreetMap (OSM) (https://www.openstreetmap.org/)[45].

monsoon season[43]. Dhaka located in a geographical position, has to faces some extreme natural disasters like floods, storms, and heat wave[44].

Data sources and methods

Multiple remotely sensed datasets were used to assess the UELI of Dhaka for 2004, 2014, and 2024 (Table 1). Landsat imagery was used to derive the Normalized Difference Vegetation Index (NDVI), Wetness, Index-based Built-up Index (IBI), and Land Surface Temperature (LST), while MODIS and nighttime light datasets from DMSP-OLS and VIIRS were used to derive the Air Pollution Index (API) and Night Light Index (NLI), respectively. All satellite data were processed using GEE, while R (version 4.5.2) and ArcGIS 10.8 were used for statistical analysis, spatial processing, and mapping. The datasets were subjected to preprocessing, normalization, spatial alignment, and resampling to a common 30 m grid where necessary.

The derived indicators were integrated using a data-driven analytical framework. Principal Component Analysis (PCA) was applied to NDVI, Wetness, and IBI to derive the Built Environment Index (BEI), while the Fuzzy Analytic Hierarchy Process (FAHP) was used to determine the weights of the indicators and construct the UELI. LST was further used to derive the Urban Heat Island Effect Intensity Index (UHIEII). Finally, UELI change analysis, spatial autocorrelation, and sensitivity analysis were performed to evaluate temporal changes, spatial clustering, and the robustness of the livability assessment. The overall methodological framework is presented in Fig. 2.

Extraction of remote sensing indices

NDVI

NDVI defines as a complex relationship between vegetation cover and urban ecology in order to sustain an ecological balance and develop the urban environment[46]. NDVI is used for assessing the greenness which is effectively reflects the vegetation condition of the area is great for detect the changes of land use pattern (Fig. 3). The NDVI has become the index most commonly employed for vegetation analysis due to its long record, straightforward use, and dependency on widely accessible multi-spectral bands[47]. The both positive and higher NDVI value indicates the strong vegetation cover. Meanwhile the lower value indicates less implication of vegetation coverage. The Value ranges between -1 to 1[48].

Data	Source	Dataset	Resolution	Year
NDVI	Google Earth Engine	LANDSAT/LE07/C02/T1_TOA	30 m	2004
		LANDSAT/LC08/C02/T1_TOA		2014
		LANDSAT/LC09/C02/T1_TOA		2024
Wetness		LANDSAT/LT05/C02/T1_TOA	30 m	2004
		LANDSAT/LE07/C02/T1_L2		2014
		LANDSAT/LC09/C02/T1_L2		2024
Index- based Built-up Index (IBI)		LANDSAT/LT05/C02/T1_L2	30 m	2004
		LANDSAT/LE07/C02/T1_L2		2014
		LANDSAT/LC08/C02/T1_L2		2024
The Urban Heat Island (UHI)		LANDSAT/LT05/C02/T1_L2	30 m	2004
		LANDSAT/LC08/C02/T1_L2		2014
		LANDSAT/LC08/C02/T1_L2		2024
Air Pollution Index (API)		MODIS/061/MCD19A2_GRANULES	1000 m	2004
		MODIS/061/MCD19A2_GRANULES		2014
		MODIS/061/MCD19A2_GRANULES		2024
Night Light Index (NLI)		NOAA/DMSP-OLS/NIGHTTIME_LIGHTS	1000 m	2004
		NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG	500 m	2014
		NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG		2024

Table 1. Details of the database used in Urban Livability environment index Mapping in Dhaka City.

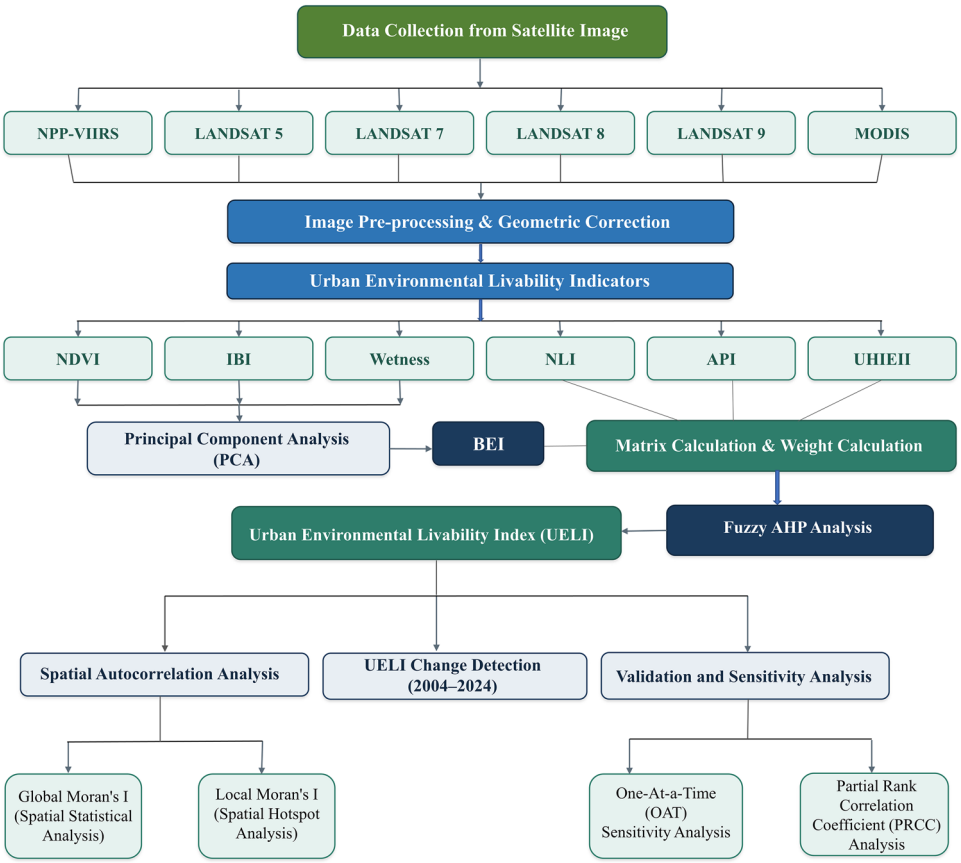


Fig. 2. Flowchart of the study methodology. The figure was prepared by the authors using Microsoft PowerPoint.

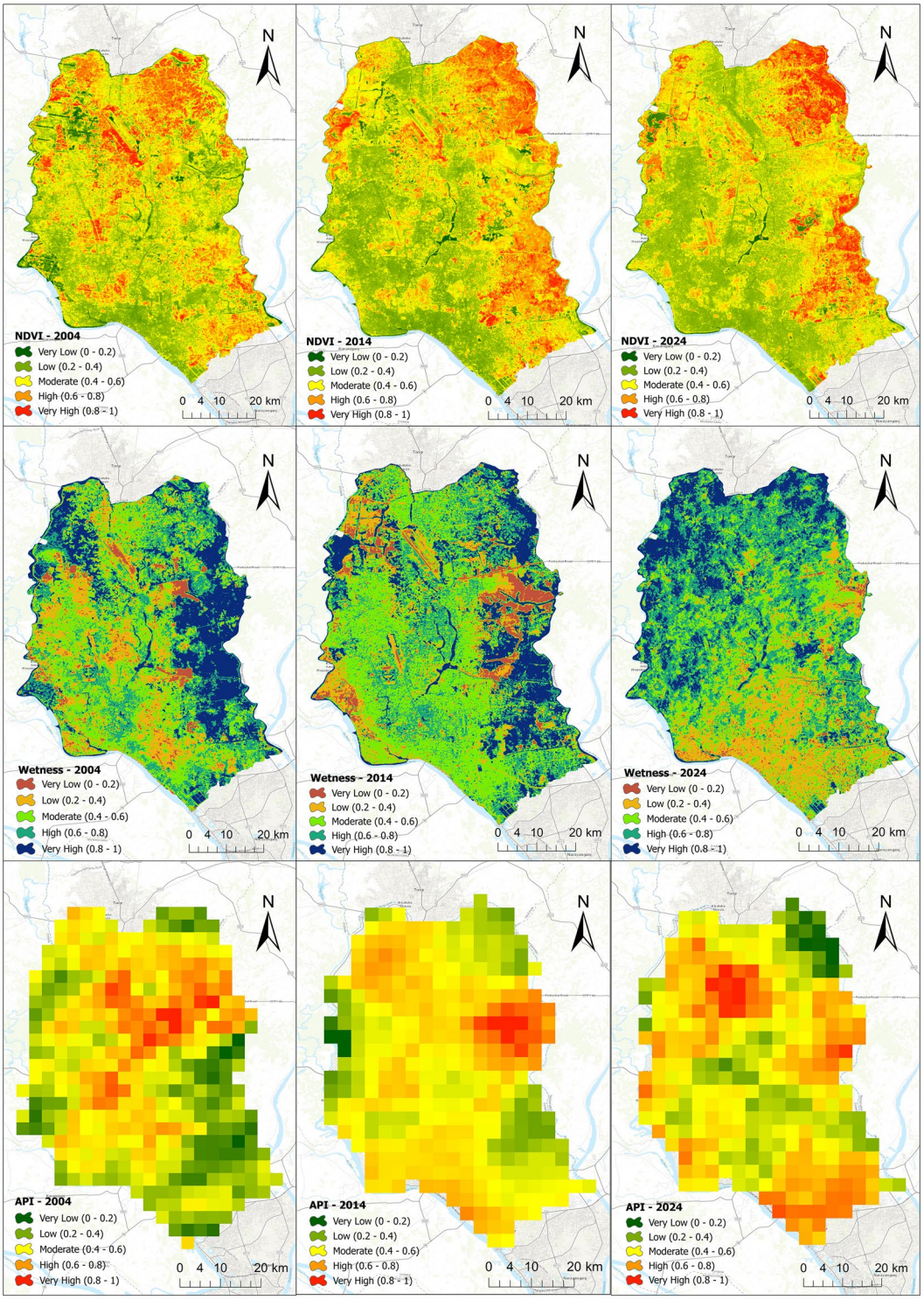


Fig. 3. NDVI, Wetness, API parameter for different years 2004, 2014 and 2024. These indices were processed and generated using Google Earth Engine (GEE) (https://earthengine.google.com/). Final map composition and layout were prepared by the authors using ArcGIS Desktop 10.8 (ArcMap), Esri (https://www.esri.com/arcgis).

$$NDVI = \frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + \rho_{RED}} \quad (1)$$

where ρ_{NIR} indicates the near—infrared band, ρ_{RED} indicates the red band as Spectral reflectance measurements which are recorded using sensors in required areas[49].

Wetness

To determine the wetness of the study area, we calculated the Tasseled Cap Transformation (TCT) technique. The Tasseled Cap Transformation (TCT), or Kauth–Thomas transformation, converts multispectral image bands into a weighted index comprising a set of components, with each pixel of the index derived from the original multispectral remote sensing data[50]. Since the TCT coefficients differ across Landsat sensors, this study used the coefficients derived for Landsat 8 OLI, Landsat 7 ETM+, and Landsat 5 at-satellite reflectance[51–53]. The TCT equation is-[54,55]

$$Y = cx + a \quad (2)$$

For Landsat 8/9 (OLI):

$$Wet = 0.1511 \cdot Blue + 0.1973 \cdot Green + 0.3283 \cdot Red + 0.3407 \cdot NIR - 0.7117 \cdot SWIR1 - 0.4559 \cdot SWIR2 \quad (3)$$

For Landsat 7 ETM+:

$$Wet = 0.2626 \cdot Blue + 0.2141 \cdot Green + 0.0926 \cdot Red + 0.0656 \cdot NIR - 0.7629 \cdot SWIR1 - 0.5388 \cdot SWIR2 \quad (4)$$

For Landsat 5 TM:

$$Wet = 0.1509 \cdot Blue + 0.1973 \cdot Green + 0.3279 \cdot Red + 0.3406 \cdot NIR - 0.7112 \cdot SWIR1 - 0.4572 \cdot SWIR2 \quad (5)$$

where *c* is the transformation coefficient, which is connected to the sensor on board a satellite, a is a constant used to prevent negative values, y is the component in multispectral space following the transformation, and x represents the multispectral bands before the transformation[56].

Index-based built-up index (IBI)

IBI is an impactful technique to indicate the urban growth level or built-up area. IBI is a complex index is generated by using the other index which obtained from the original imagery bands[57] NDBI, which uses the visible, NIR, and SWIR bands, has limitations that the IBI can get around. In addition to the NIR band, this study uses two Vis-based indices, VrNIR-BI and VgNIR-BI, which use the red and green bands, respectively[58].

$$UI = \frac{\rho_{SWIR2} - \rho_{NIR}}{\rho_{SWIR2} + \rho_{NIR}} \quad (6)$$

$$NDBI = \frac{\rho_{SWIR1} - \rho_{NIR}}{\rho_{SWIR1} + \rho_{NIR}} \quad (7)$$

$$IBI = \frac{\frac{2\rho_{SWIR1}}{\rho_{SWIR1}+\rho_{NIR}} - \left[\frac{\rho_{SWIR1}-\rho_{NIR}}{\rho_{SWIR1}+\rho_{NIR}}\right]}{\frac{2\rho_{SWIR1}}{\rho_{SWIR1}+\rho_{NIR}} + \left[\frac{\rho_{SWIR1}-\rho_{NIR}}{\rho_{SWIR1}+\rho_{NIR}}\right]} \quad (8)$$

$$VrNIR - BI = \frac{\rho_{Red} - \rho_{NIR}}{\rho_{Red} + \rho_{NIR}} \quad (9)$$

$$VgNIR - BI = \frac{\rho_{Green} - \rho_{NIR}}{\rho_{Green} + \rho_{NIR}} \quad (10)$$

Built-up environment index (BEI)

BEI is calculated using NDVI, Wet, and IBI indices which are derived from Landsat-8 data with less than 10% cloud cover throughout a year are chosen for their average values. Because these indices have intricate relationships, PCA is used to reduce dimensionality. PCA is used to minimize dimension, considering the complex interactions between these indices. This method eliminates the requirement for subjective indicator weighting by enabling the extraction of significant information[59]. BEI represents the degree of urbanization, vegetation cover, and humidity within the urban environment accurately.

$$BEI_0 = PC1[f(Wet, NDVI, IBI)] \quad (11)$$

$$BEI = \frac{BEI_0 - BEI_{0min}}{BEI_{0max} + BEI_{0min}} \quad (12)$$

In Eq. (11) the three indices- Wet, NDVI, & IBI is the normalized value of the indicators. The first principal component (PC1) is selected for assessing the built environment condition by calculating BEI_0. In Eq. (12), BEI represents the final normalization value from 0 to 1 by calculating with the maximum value (BEI_{0max}) and minimum value (BEI_{0min}) of BEI_0.

The urban heat island (UHI)

The urban heat island (UHI) effect is determined as one of the adverse impacts of urbanization on the environment. Both satellite LST data and air temperature measurements can be used to assess the UHI effect[60]. This study employs the Landsat dataset to calculate Land Surface Temperature (LST) deploying inversion on the GEE platform since the UHI effect is constant around the year. Normalizing the mean surface temperature value to calculate the UHI effect's strength.

$$LST = \frac{LST_{Day} + LST_{Night}}{2} \tag{13}$$

$$UHIEII = \frac{LST - LST_{Mean}}{LST_{Mean}} \tag{14}$$

In Eq. (13) Both LST_{Day}& LST_{Night} represents the average daytime and nighttime temperature derived from the Landsat image for calculating the LST of the study[61]. In Eq. (14) The average LST which is LST_{Mean} value is used for calculating the Urban Heat Island intensity.

Air pollution index (API)

API is a widely recognized measure of the health effects of air quality, was developed to evaluate the cleanliness and quality of the air by reporting pollutant levels. An indicator of clean air, API is a parameter that should be kept as low as feasible[32]. To conduct the API, Aerosol Optical Depth (AOD) derived from the MODIS Collection 6.1 MAIAC product, a combined of Terra and Aqua dataset with 1 km spatial resolution. A linear regression model used for estimating PM2.5 from AOD.

$$\mathrm{PM2.5} = \mathrm{m} \times \mathrm{AOD} + \mathrm{c} \tag{15}$$

Here, m and c represent the slope and intercept were derived from the study area[62]. AOD considered as a proxy for gaseous particulate pollutant and such as NO_2, CO, or O_3.

Night light index (NLI)

The NTL images, which are generated by detected light radiation caused by human activity on the earth's surface, can directly reflect artificial surface regions and areas with significant human activity[63,64]. NTL remote sensing data provides continuous and reasonably accurate global nighttime brightness information in order to determine human economic activity, infrastructure levels, urbanization, and urban built-up areas[65,66]. There are two main nighttime light datasets: the Visible Infrared Imaging Radiometer Suite (VIIRS) and the Defense Meteorological Satellite Program–Operational Linescan System (DMSP-OLS), which has data from 1992 to 2013 and a resolution of about one kilometer[67]. In this research, annual NPP-VIIRS nighttime light (NTL) images (Fig. 4) were extracted from GEE to analyze urban expansion dynamics of Dhaka.

Quantitative techniques

Principal component analysis (PCA)

PCA is a form of statistical analysis based on reducing dimensionality that makes it possible to reduce a large number of variables to a small collection of orthogonal principal components[68]. The fundamental features included in a complicated multivariate dataset can be logically and systematically analyzed through the geometric properties of the PCs. PCA evaluates for the indicators' linear combination with the highest variance[69]. PCA is a robust, data-driven technique with no assumptions about limitations. This implies that there is no automatic satisfactory or undesirable test for value; instead, researchers must apply their own knowledge to determine the number of components that truly matter[69]. As previously noted, the evaluation first normalizes these individual indicators to make them dimensionless and all within the range of [0, 1] in order to eliminate the effects brought about by data measurement units and varied ranges of the indicators. The correlation coefficient matrix's eigenvalues and orthonormal eigenvectors for the dimensionless data matrix will next be determined[70]. The normalization is performed using the following min–max transformation:

$$NLI = \frac{I_0 - I_{min}}{I_{max} - I_{min}} \tag{16}$$

where (I_0) is the initial value of the indicator, and I_{max} and I_{min} represent the maximum and minimum values of that indicator, respectively[71].

By standardizing all indicators to the same range, this normalization technique guarantees that the contributions of various indicators to the principal components remain on a consistent scale during the next PCA analysis.

PC1 is then calculated via PCA:

$$PC1 = \int (Wet, NDVI, IBI) \tag{17}$$

The first principal component (PC1) is selected for assessing the normalized value of NDVI, WET and IBI to calculate the Built-up Environment Index (BEI).

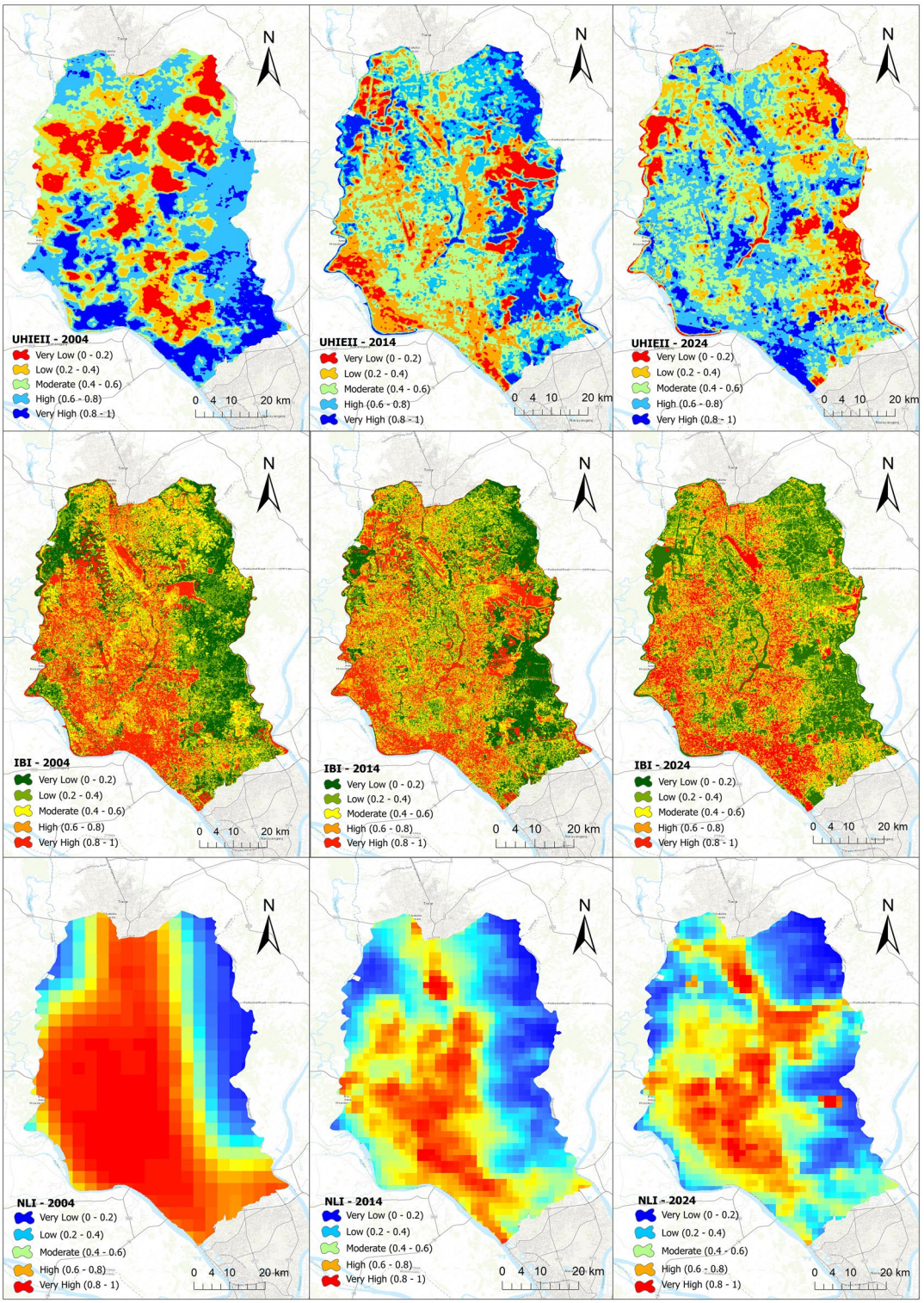


Fig. 4. UHIEII, IBI, NLI parameters from different years 2004, 2014, and 2024. These indices were processed and generated using Google Earth Engine (GEE) (https://earthengine.google.com/). Final map composition and layout were prepared by the authors using ArcGIS Desktop 10.8 (ArcMap), Esri (https://www.esri.com/arcgis).

Fuzzy analytic hierarchy process (FAHP)

FAHP employed in decision evaluation to assist those making decisions in reaching logical conclusions in challenging multi-criteria decision problems[72]. FAHP is the variation of Analytic Hierarchy Process (AHP) which is an approach to system analysis that combines qualitative and quantitative methods[73,74]. According

Saaty level	Explanation	Triangular fuzzy number
1	Equally important	(1, 1, 1)
3	Slightly important	(2, 3, 4)
5	Quite important	(4, 5, 6)
7	Very important	(6, 7, 8)
9	Absolutely important	(9, 9, 9)
2, 4, 6, 8	Interval value between neighboring scales	(1, 2, 3), (3, 4, 5), (5, 6, 7), (7, 8, 9)

Table 2. Evaluation of Linguistic terms and TFNs.

Index	BEI	API	NLI	UHIEII
BEI	(1, 1, 1)	(2, 3, 4)	(2, 3, 4)	(4, 5, 6)
API	(1/4, 1/3, 1/2)	(1, 1, 1)	(1, 1, 1)	(1, 1, 2)
NLI	(1/4, 1/3, 1/2)	(1, 1, 1)	(1, 1, 1)	(1, 1, 1)
UHIEII	(1/6, 1/5, 1/4)	(1/2, 1, 1)	(1, 1, 1)	(1, 1, 1)

Table 3. Interactor judgment fuzzy number matrix.

to the article "Quantitative Fuzzy Semantics," fuzzy set theory, which has its roots in electrical engineering, offers a mathematical framework for managing the imprecision of human reasoning by permitting elements to have progressive degrees of membership through a membership function[75]. By creating hierarchical structures, establishing guidelines and sub-criteria, and utilizing fuzzy language to explain the relative importance and priority among the guidelines, fuzzy mathematics and hierarchical analysis are combined to address fuzzy and uncertain decision-making factors based on expert judgment[76,77]. FAHP uses linguistic variables, such as triangular fuzzy numbers, to make pairwise comparisons of criteria and alternatives[7].

1. Establishment of the hierarchy: From the objective layer to the indicators and sub-indicators, the decision problem was organized into a hierarchical framework. Evaluating the livability of urban environments is the goal. The sub-indicator layer corresponds to BEI, API, UHIEII, and NLI, respectively, while the indicator layer comprises ecological environment quality, urban heat island effect, and socio-economic effect (Table 2).

2. Determining indicator-level weights: To account for the imprecision of human judgment, experts used fuzzy language to describe the significance of each indicator and sub-indicator in relation to its higher-level criterion.

These judgments are represented as Triangular Fuzzy Numbers (TFN), denoted as:

$$M = (l, m, u) \tag{18}$$

where$l, m, andn$ represents the lower, modal, and upper bounds of the fuzzy sets, respectively. The function $\mu M(x)$ defined as:

$$\mu M(x) = \begin{cases} \dfrac{x-l}{m-l} l \le x \le m \\ \dfrac{u-x}{u-m} m \le x \le u \\ 0 otherwise \end{cases} \tag{19}$$

This step highlights the importance of expert knowledge in determining weight.

3. Building the matrix for fuzzy judgment: For the sub-indicators (BEI, API, UHIEII, and NLI), a judgment matrix based on fuzzy comparisons was developed, capturing pairwise preferences among them using linguistic variables converted into fuzzy numbers (Table 3).

4) Calculating the weight vector: The weight vector, which represents the relative importance of each sub-indicator, was calculated by applying fuzzy mathematical operations to the judgment matrix. For instance, the weights for BEI, API, UHIEII, and NLI were determined to be 0.30, 0.25, 0.20, and 0.25, respectively.

(5) Testing consistency: To verify the judgment matrix, the consistency ratio (CR) was calculated. Acceptable consistency is indicated by a CR < 0.1. CR = 0.04 in this investigation indicates that the matrix is consistent.

$$CI = \frac{\lambda_{max} - n}{n-1} \tag{20}$$

$$CR = \frac{CI}{RI} \tag{21}$$

(6) Extensive evaluation and normalization: The UELI was calculated by applying the final weights to the normalized indices, resulting in a composite map that spatially explicitly represents urban environmental quality.

Spatial auto correlation analysis

The principle of Spatial autocorrelation contributes to identifying the spatial analysis employing spatial statistics and spatial econometrics on its contents[78]. Metrics of spatial autocorrelation can be defined by applying a globally normalized weight matrix and a normalized vector[79]. Spatial correlation is a very helpful tool for finding patterns of social phenomena that take place in space when combined with typical exploratory data analysis[80].

Global Moran's I analysis

Global Moran's I is the primary metric used to assess global spatial autocorrelation, which describes the attribute values in the spatial features of the entire region[81].

This approach takes into account both feature locations and attribute values simultaneously to assess whether the observed spatial distribution is clustered, scattered, or random[82,83].

Global measurements (I) use a single number that encapsulates the average behavior of the data to characterize the autocorrelation across the entire study area.

$$I = \frac{N\sum_{i=1}^{n}\sum_{j=1}^{n}W_{ij}\left(x_i-\overline{x}\right)\left(x_j-\overline{x}\right)}{\left(\sum_{i=1}^{n}\sum_{j=1}^{n}W_{ij}\right)\sum_{i=1}^{n}\left(x_i-\overline{x}\right)^2} \tag{22}$$

Here, N is the total number of spatial units. x_i and x_j are the attribute values for units and j. $\overline{x}$ is the mean value of the attribute across all units. W_{ij} represents the elements of the spatial weight matrix W, which defines the connectivity or distance relationship between units and j[84]

Moran's I analyzed the critical statistical value by using the z-score and p-value. The calculation of the z-score:

$$Z = \frac{I - E[I]}{\sqrt{VAR[I]}} \tag{23}$$

$$E[I] = \frac{-1}{(n-1)} \tag{24}$$

Here, $E[I]$ is the estimated expected value of spatial randomness, where $p < 0.05$ signifies a high z-score value[85]. The value of Moran's I varies between -1 and + 1. The spatial clustering of specified characteristics is shown by a positive value, whereas a negative number indicates regional dispersion (high-low or low–high)[86].

LISA cluster analysis

An essential tool for exploratory spatial analysis is LISA (Local Indicators of Spatial Association) analysis, which makes it feasible to visualize and examine links between regions.

Local measures are well suited to three basic aspects: (a) identification of the existence of hot spots, (b) assessing the assumptions of stationarity, and (c) identification of threshold distances beyond which no discernible spatial association exists[87]

In this research, LISA was performed on a regular grid created by aggregating the UELI raster (original resolution 30 m) by a factor of 10, resulting in a final grid cell size of 300 × 300 m (0.3 × 0.3 km). Spatial relationships were defined using a first-order queen contiguity neighborhood with a row-standardized weight matrix, and classified by significance at $p \leq 0.05$. The Queen technique is useful because it takes into account the greatest number of neighboring areas that share corners or edges, demonstrating more adjacency and enabling a more in-depth examination of regional linkages[88]. The technique is invaluable for examining geographical correlations, like urban livability, and makes the spatial link between locations clear. The following is an expression for the weighted regression equation:

$$L_{it} = \alpha_i + \beta_1 W_{it} + \beta_2 X_{it} + \varepsilon_{it} \tag{25}$$

1. L_{it} is the livability index for area i at time t
2. α_i represents the intercept specific to each area
3. W_{it} is the spatial weight matrix derived from the Queen contiguity method
4. X_{it} is a vector of independent variables affecting livability
5. ε_{it} is the error term

High & High (HH) clusters, High & Low (HL) clusters, Low & High (LH) clusters, and Low & Low (LL) clusters are the four categories into which the LISA analysis outcomes are divided. LL clusters are core areas of low livability, while HH clusters can be categorized as core areas of high livability. While LH and HL clusters can be thought of as spatial outlier regions, the spatial clusters mostly relate to the HH and LL kinds[89].

Results

PCA analysis

Complex data can be made much simpler through PCA. It assists in rendering complex signs easier to comprehend and work with by assisting us in identifying their key components. We used PCA to examine the BEI for Dhaka City in 2004, 2014, and 2024 (Fig. 5). A harmonious combination of NDVI For describing the vegetation cover and stresses, Wetness for indicating the humidity, IBI to characterize a high-quality built environment of the study area.

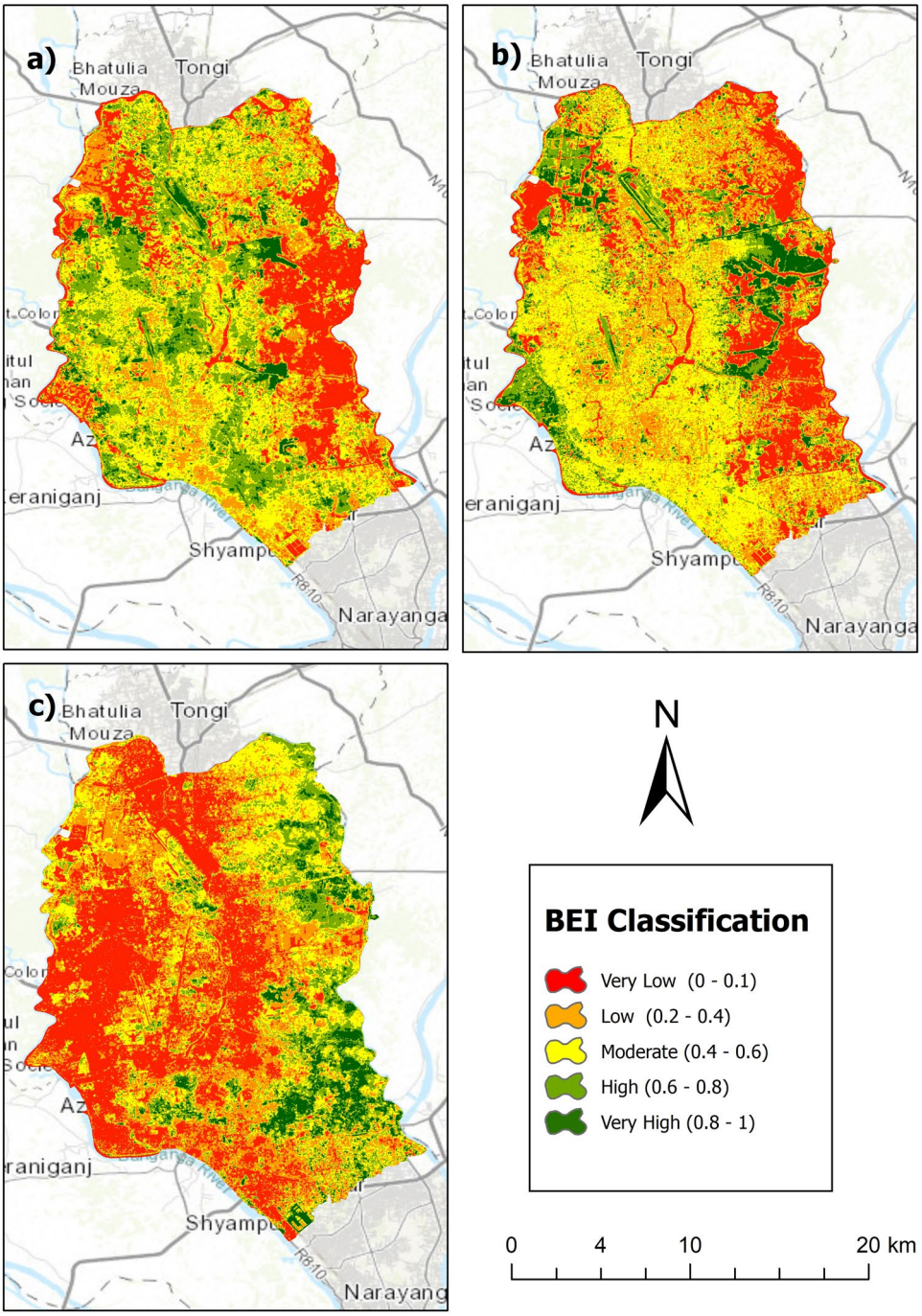


Fig. 5. BEI classification maps for (**a**) 2004, (**b**) 2014, and (**c**) 2024. BEI was derived using principal component analysis (PCA) of NDVI, Wetness, and IBI, and the final maps were laid out using ArcGIS Desktop 10.8 (ArcMap), Esri (https://www.esri.com/arcgis).

In the PCA, the NDVI became the significant variable with a strong negative value on PC2 (-0.96) in 2004, went to PC3 (-1.00) in 2014 and the dataset variance reached on PC1 (-0.99) in 2024 which marked a substantial shift in the order of influence. The IBI became the dominated factor for PC3 (1.00) in 2004, which became a defining characteristics for PC2 (-1.00) in 2014 and again tried to play an important variable for PC (0.99) in 2024. Wetness, which had a high positive value on PC1 (0.96) in 2004, continued to kept the consistency with strong positive value on PC1 (1.00) & reached to dominated value on PC2 (0.99) in 2024 compared to other variables (Table 4).

The indices maintained extremely high loadings (around 1.0 or -1.0) over the course of all three years, indicating that although the rank of the components changed, the variables themselves stayed separate and did not considerably overlap in the environment.

Year	Indicators/Statistics	PC1	PC2	PC3
2004	NDVI	−0.25	−0.96	0.00
	IBI	−0.00	0.00	1.00
	Wetness	0.96	−0.25	0.00
	Eigenvalue	0.13	0.09	0.00
	Contribution Rate (%)	65.21%	34.78%	0.01%
2014	NDVI	0.00	−0.00	−1.00
	IBI	−0.00	−1.00	0.00
	Wetness	1.00	−0.00	0.00
	Eigenvalue	0.08	0.00	0.00
	Contribution Rate (%)	89.52%	10.46%	0.02%
2024	NDVI	−0.9954	−0.0957	−0.0001
	IBI	−0.00	0.00	0.99
	Wetness	−0.09	0.99	−0.00
	Eigenvalue	0.14	0.05	0.00
	Contribution Rate (%)	87.77%	12.22%	0.01%

Table 4. Different variables distribution on principal components in 2004, 2014 and 2024 respectively.

In 2004, 2014, and 2024, the first principal component's eigenvalue contribution rates were 65.21%, 89.52%, and 87.77%, respectively, all of which were higher than 60%. This suggests that most of the data from the three indicators is collected by the first main component. As a result, PC1 was employed in this study as the environmental livability assessment factor for indicators of the quality of the urban built environment.

It indicates that, from 2004 and 2024, Dhaka's eastern regions Uttara, Badda, Khilgaon, Bonosri, Demra had the best environmental quality. The western and the core urban centers (Motijheel, Ramna, Kotowali, Lalbag, Hazaribag, Tejgaon, Mohammodpur, Hazaribagh, Kamraingichar, Gulshan, Banani) the quality of the built environment has sadly declined. Positively, the quality of the environment in Mirpur and Cantonment area has improved.

Fuzzy analytic hierarchy process (FAHP)

The quality of the built environment is described by the BEI, whereas the quality of the air environment is represented by the API. When taken as a whole, these two metrics show the state of the ecological environment. While the UHIEII measures urban thermal stress, the NLI is a stand-in for economic development. A comprehensive environmental livability index was created by applying the FAHP across several spatial scales in order to integrate these indicators. The normalized weights derived from FAHP were 0.30 for BEI, 0.25 for NLI, and 0.20 for UHIEII and 0.25 for API. With a consistency ratio of 0.04, the judgment consistency was deemed acceptable. Additionally, the equal interval classification method was used to distinguish different levels of environmental livability based on expert knowledge and the statistical distribution of the data (Fig. 6).

Between 2004 and 2024, Dhaka's high to very high heat intensity spread steadily, according to UHIEII maps. Due to rapid urbanization and the loss of ecological buffers, lower-intensity zones have progressively moved to higher levels, particularly in the eastern and central regions. The API maps show that between 2004 and 2024, Dhaka's air quality steadily decreased, with high and extremely high pollution zones expanding (Fig. 7). The western and central parts of the city are seeing an increase in the concentration of these degraded areas, which is indicative of higher emissions linked to traffic and urban growth. As well as the NLI values in central part of Dhaka increased significantly from 2004 to 2021. This index reveals a significant change between 2004 to 2024. According to the UELI maps, between 2004 and 2024, the central, western, and southern regions of Dhaka saw an increasing concentration of high and very high livability intensity. meanwhile, the eastern and outermost regions continue to be dominated by low to very low UELI zones, indicating distinct geographical differences in urban livability.

In ArcGIS 10.8 software, we have reclassified using the spatial analyst tool and categorized the UELI value of Dhaka area into four livable conditions. The statistical calculation of area for each class were extracted from the raster attribute table in ArcMap. The proportion of each livability class was then calculated based on pixel area. In the calculation, the percentage of unlivable areas was 15.65% in 2004, 17.84% in 2014, and 30.80% in 2024. The proportion of less livable locations was 36.03% in 2004, 39.38% in 2014, and 35.51% in 2024. In contrast, the proportion of livable areas was 30.53% in 2004, 26.87% in 2014, and 20.75% in 2024 (Table 5). Conversely, very livable areas accounted for only 17.78% of the total area in 2004, 15.89% in 2014, and 12.93% in 2024. The findings indicate that less livable conditions predominated in the study area throughout all years, with a slight decrease in livable areas by 2024, same as the extremely livable conditions remained decreasing. The study area is relatively appropriate for human occupancy, according to the findings. Though less livable places predominate throughout the research period. The very livable category is still quite small, though, and the rise in unlivable places in 2024 suggests that environmental or infrastructure stress is getting worse. Although the land can support occupancy overall, sustainable living conditions require better planning, environmental management, and infrastructure development (Tables 6, 7, 8).

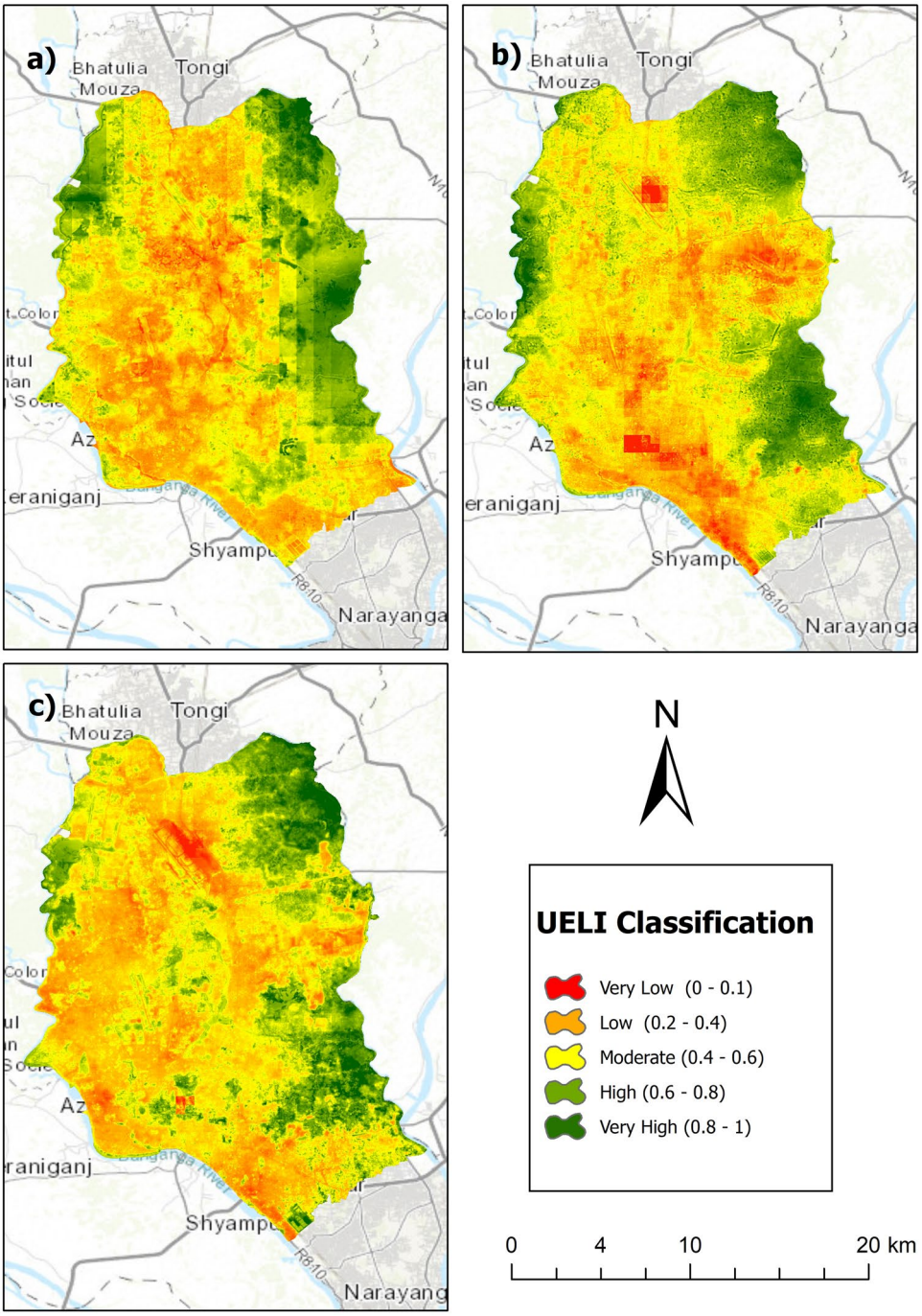


Fig. 6. UELI classification maps for (**a**) 2004, (**b**) 2014, and (**c**) 2024. Maps generated using ArcGIS Desktop 10.8 (ArcMap), Esri (https://www.esri.com/arcgis).

UELI change analysis

The UELI transition matrices between 2004–2014, 2014–2024, and 2004–2024 represent the accelerating degradation in urban environmental livability, which is consistent with unplanned eastern peri-urban expansion patterns.

Less Livable—Less Livable remained the dominant class from 17.33% to 17.66% in 2004 -2014 and 2014–2024, which is cumulatively 15.27%. It reflects a stable but degraded urban core, while Very Livable—Very Livable grew modestly from 7.58% to 8.93% in 2004–2014 and 2014–2024, which indicates localized consolidation of remaining green or riverine tracts. Less Livable—Unlivable shows the clearest surge signal in negative transition, doubling from 6.89% to 15.22% in 2004 -2014 and 2014 -2024, and cumulatively 13.96% between 2004–2024. Similarly, Livable—Unlivable rose from 4.65% to 5.84% and 8.70% cumulatively. Thus, the upgrading transitions from Unlivable—Very Livable are from 0.21% to 0.30%, which stayed marginal throughout and confirms that

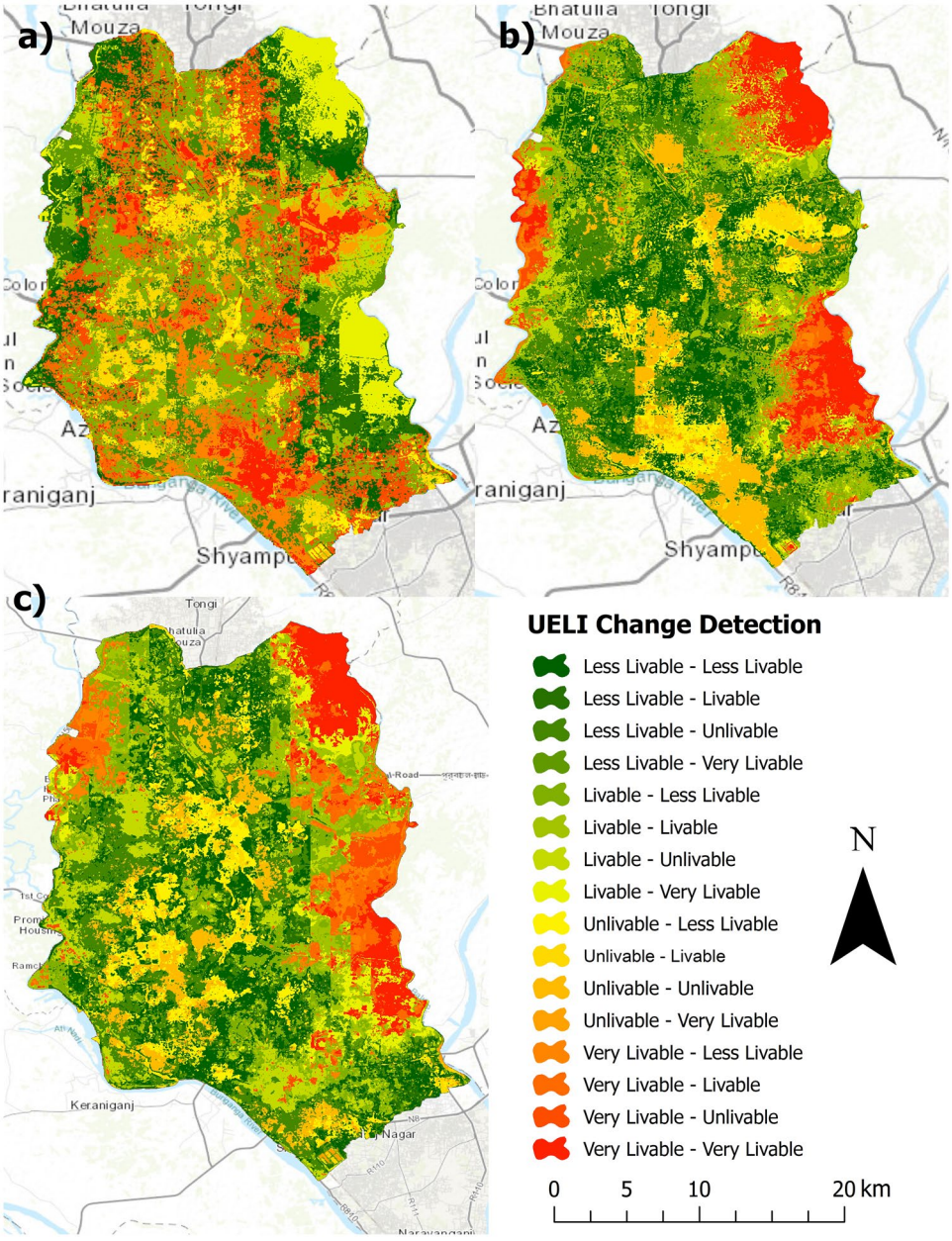


Fig. 7. UELI change detection maps for (**a**) 2004–2014, (**b**) 2014–2024, and (**c**) 2004–2024. Maps generated using ArcGIS Desktop 10.8 (ArcMap), Esri (https://www.esri.com/arcgis).

Class	2004		2014		2024	
	Area (km^2)	Percentage (%)	Area (km^2)	Percentage (%)	Area (km^2)	Percentage (%)
Unlivable	47.77	15.65	54.42	17.84	93.93	30.80
Less livable	109.99	36.03	120.11	39.38	108.29	35.51
Livable	93.18	30.53	81.69	26.87	63.29	20.75
Very livable	54.29	17.78	48.47	15.89	39.44	12.93

Table 5. Statistical analysis of the livability of the urban environment by area.

livability loss structurally outpaces gain, and it aligns with nighttime-light evidence of unplanned, inequality-driven urban growth concentrated in Dhaka's expanding fringe.

The dark-green Less livable persistence dominates the central-western core across all periods, while the eastern and northeastern fringe shifts from green-yellow from 2004 to 2014, toward orange-red from 2014

Transition category	2004 to 2014 (km^2)	Percentage (%)	2014 to 2024 (km^2)	Percentage (%)	2004 to 2024 (km^2)	Percentage (%)
Very Livable—Very Livable	23.10	7.58	27.23	8.93	21.02	6.89
Very Livable—Livable	14.72	4.83	13.73	4.50	16.42	5.38
Very Livable—Less Livable	11.58	3.80	4.96	1.63	10.64	3.49
Very Livable—Unlivable	4.88	1.60	2.52	0.83	6.20	2.03
Livable—Very Livable	19.13	6.27	8.85	2.90	12.88	4.22
Livable—Livable	29.09	9.54	25.05	8.21	22.99	7.54
Livable—Less Livable	30.76	10.09	30.25	9.92	30.74	10.08
Livable—Unlivable	14.17	4.65	17.80	5.84	26.54	8.70
Less Livable—Very Livable	5.58	1.83	2.54	0.83	4.50	1.48
Less Livable—Livable	30.36	9.96	17.29	5.67	16.18	5.31
Less Livable—Less Livable	52.86	17.33	53.86	17.66	46.57	15.27
Less Livable—Unlivable	21.01	6.89	46.40	15.22	42.56	13.96
Unlivable—Very Livable	0.65	0.21	0.90	0.30	1.12	0.37
Unlivable—Livable	7.78	2.55	7.16	2.35	7.66	2.51
Unlivable—Less Livable	24.90	8.16	19.16	6.28	20.29	6.65
Unlivable—Unlivable	14.39	4.72	27.22	8.93	18.66	6.12

Table 6. UELI change detection analysis for 2004–2014, 2014–2024, and 2004–2024.

	2004	2014	2024
Observed Moran's Index	0.9543	0.9545	0.9379
Expected Index	-2.94×10^{-05}	-2.94×10^{-05}	-2.94×10^{-05}
Variance	1.29×10^{-05}	1.29×10^{-05}	1.29×10^{-05}
Z-score	265.49	265.47	260.86
p-value	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$	$< 2.2 \times 10^{-16}$

Table 7. Statistical analysis of Global Moran's I spatial autocorrelation for the livability of the urban environment.

	2004			2014			2024		
Cluster category	Count	Z Score	*P* Value	Count	Z Score	*P* Value	Count	Z Score	*P* Value
High-High	6103	2.60	0.017	5211	3.05	0.012	5016	2.64	0.018
High-Low	1	−2.47	0.014	1	−2.01	0.044	4	−2.04	0.041
Low–High	1	−1.96	0.049	1	−2.01	0.043	1	−2.00	0.045
Low-Low	6744	3.29	0.009	6226	3.31	0.008	6339	3.43	0.008
Not significant	21,165	0.91	0.405	22,556	0.8	0.454	22,634	0.88	0.411

Table 8. Statistical analysis of the LISA cluster map.

to 2024, cumulatively, marking a rapid transition from livable to unlivable cities. This eastern hotspot aligns with active horizontal expansion, consistent with growth models linking densification to declining biophysical indicators such as NDVI and wetness. The red Very Livable patches representing the eastern, northeastern, and southern areas are likely to be protected. Overall, degradation intensified in the second decade, concentrated along the eastern fringe, underscoring the need for targeted intervention there to arrest further livability loss.

Global Moran's I analysis

In this study, the UELI was analyzed by the Global Moran's I to identify the statistically significant pattern of spatial autocorrelation among the variables for 2004, 2014 & 2024 (Fig. 8). This analysis tried to indicate the same clustered area value through the single Moran's Index with five categorical values: the Observed Moran's Index, Expected Index, Variance, z-score, and *p*-value.

For 2004, the Moran's I value is 0.9543 with which is > 0 indicate a random cluster pattern with consistent positive statistical value that close to zero ($p < 0.0001$). The positive z-score value is 265.49 indicates the UELI dataset is more clustered and processed in a random way. In 2014, the Moran's I value identifies as the highest value which is 0.9545. The p-value approaching to the zero with highest z-score value 265.47 which significantly include the categorical rejection of the null hypothesis of spatial randomness. But in 2024, the Moran's I value

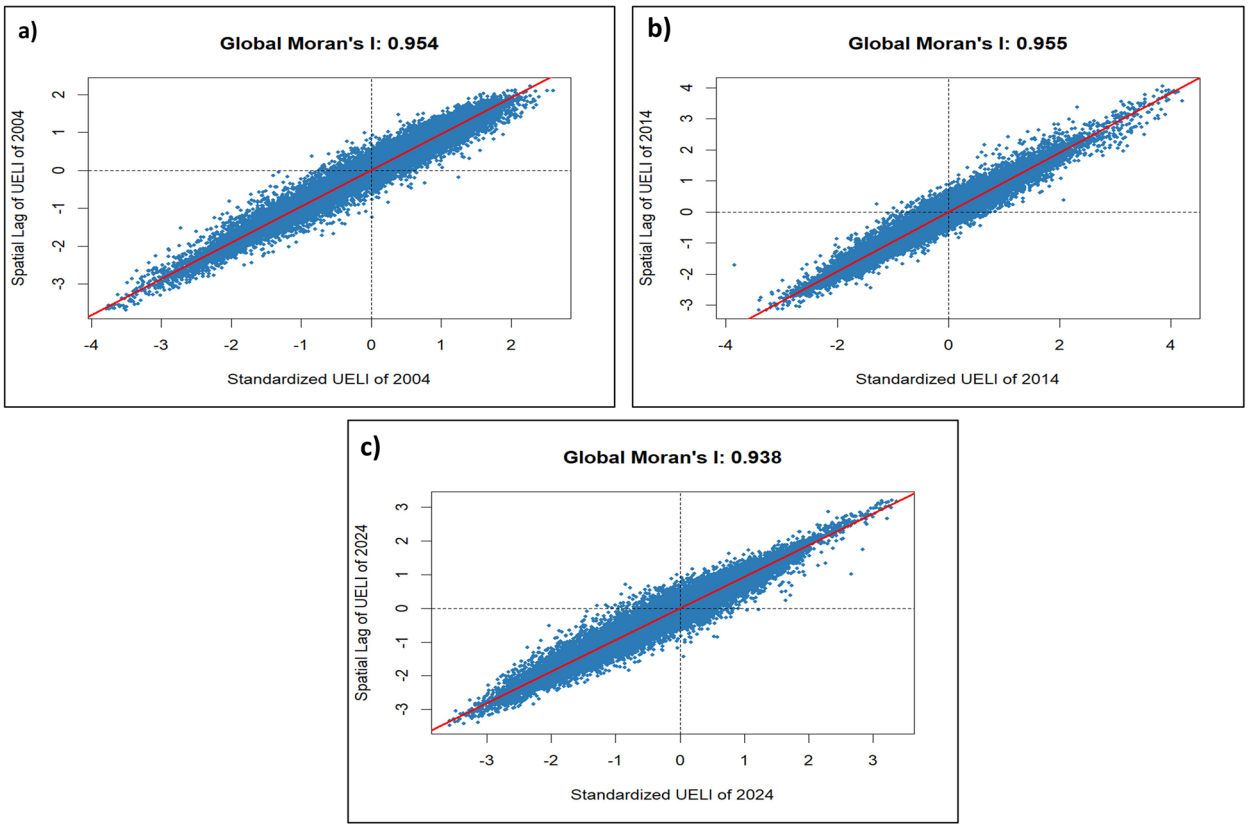


Fig. 8. Moran's I scatter plot for UELI in 2004, 2014 and 2024. These plots were generated using R software (version 4.5.2).

is slightly declining and settling at 0.9379. The z-score found to be 260.86 which is applied to identify the data variance of standard deviation in dataset.

The scatter plot is used in spatial statistics to measure the Global Moran's I across three different years: 2004, 2014, and 2024 in a global trend. The scatter plot is categorized into four different quadrants (HH, HL, LH, and LL) to represent the spatial relationship. The upper right quadrant represents the High-High (HH) values, which signify the hot spot of high values, and the upper left for Low–High (LH) values. The lower right for High-Low (HL) values, and the lower left for Low-Low (LL) values, are determined as the cold spots surrounded by the low categorical values. The regression line represents Moran's I coefficient to indicate the spatial autocorrelation of UELI. The steep positive slope of the regression line determined a strong neighborhood effect with high UELI values.

LISA cluster analysis

Local Indicators of Spatial Association (LISA) is using for illustrate the spatial autocorrelation of a specific variable in a specific region. Here, LISA is employed to identify the map areas where data values have a high positive or negative correlation with one another. The livability index's global autocorrelation analysis showed a Moran's I value, which denotes a very high degree of autocorrelation on a global scale. LISA maps show outliers (High-Low or Low–High) and clusters where similar values are gathered together (High-High or Low-Low) by calculating the UELI map.

The LISA cluster maps from 2004, 2014, and 2024 show a distinct and growing spatial polarization in the area over 20 years. A large portion of the map was statistically non-significant in 2004 due to the spatial pattern's relative fragmentation, which was typified by small, isolated High-High (red) clusters in the center-south and a concentrated Low-Low (dark green) block in the northeast (Fig. 9). By 2014, a considerable change had taken place, with the Low-Low clusters consolidating into a large vertical belt along the eastern boundary and the High-High clusters expanding significantly and merging into a dominant corridor through the central and southwestern sectors. This tendency peaked in 2024, revealing a sharply divided terrain with persistent Low-Low cold areas in the east and solidified High-High hotspots dominating the western and central regions.

The general shift from 2004 to 2024 shows a strengthening of distinct geographic regimes and a decrease in random spatial distribution, which is probably due to the continuance of rural or undeveloped area in the east and the rapid urban expansion in the west.

According to the LISA analysis, the study region is continuously dominated by non-significant (NS) areas, which accounted for 193.0 km^2 (2.23%) in 2004 205.6 km^2 (66.35%) in 2014, and 206.4 km^2 (66.6%) in 2024 (Table 9). The High-High (HH) clusters decrease gradually in all two further years. Their spatial extent drops

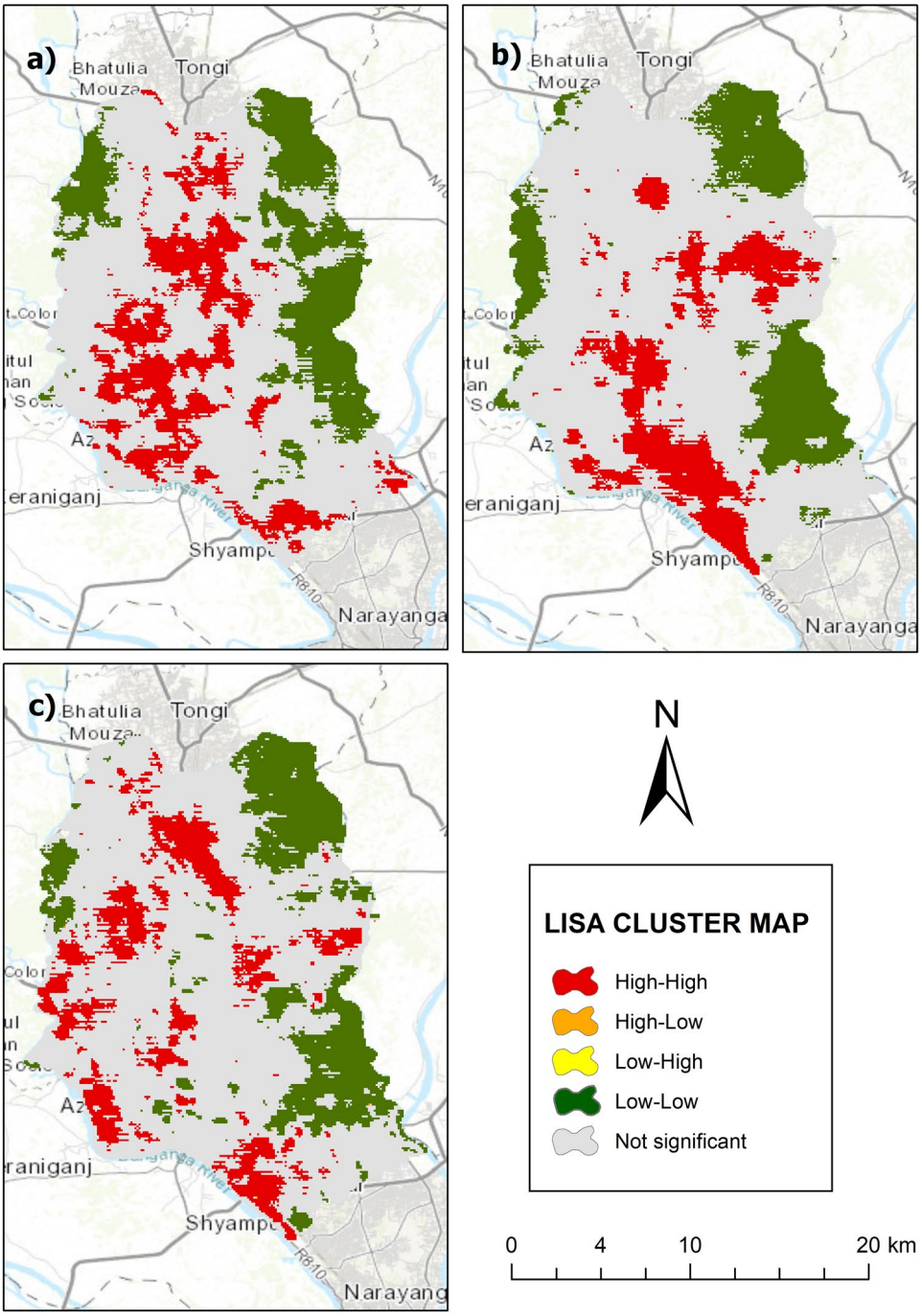


Fig. 9. LISA Cluster classification maps for (**a**) 2004, (**b**) 2014, and (**c**) 2024. These maps were generated using R software (version 4.5.2) and then layout performed in ArcGIS Desktop 10.8 (ArcMap), Esri (https://www.esri.com/arcgis).

by 8.1 km^2 from 55.6 km^2 (17.94%) in 2004 to 47.5 km^2 (15.33%) in 2014. But in 2024, HH clusters decrease to 45.7 km^2 (14.8%), which represents a net decrease of 1.8 km^2 from 2014. Low-Low (LL) clusters exhibit the opposite trend. They grow to 57.8 km^2 (18.6%) in 2024 after declining from 61.5 km^2 (19.82%) in 2004 to 56.7 km^2 (18.31%) in 2014. This shows an increase in cold-spot dominance from 2014 to 2024 of 1.1 km^2 (Table 9). Over the course of the investigation, spatial outlier clusters are insignificant. High-Low (HL) clusters remain the same from 0.01 km^2 (0.003%) in 2004 to 0.01 km^2 (0.003%) in 2014, then slightly increase to 0.04 km^2 (0.01%) in 2024. In a similar vein, Low–High (LH) clusters remain same from 0.01 km^2 (0.003%) in 2004 to 0.01 km^2 (0.003%) in 2014 and stay the same in 2024. Overall, the findings point to a weakening of spatial clustering strength with time, with a shift from a stronger hot-spot structure to increasing cold-spot and non-significant dominance in 2024.

Cluster type	2004		2014		2024	
	Area (km^2)	Percentage (%)	Area (km^2)	Percentage (%)	Area (km^2)	Percentage (%)
High–High	55.6	17.94	47.5	15.33	45.7	14.8
High–Low	0.01	0.003	0.01	0.003	0.04	0.01
Low–High	0.01	0.003	0.01	0.003	0.01	0.003
Low–Low	61.5	19.82	56.7	18.31	57.8	18.6
Not Significant	193.0	62.23	205.6	66.35	206.4	66.6

Table 9. Cluster analysis by LISA of the livability of the urban environment by area.

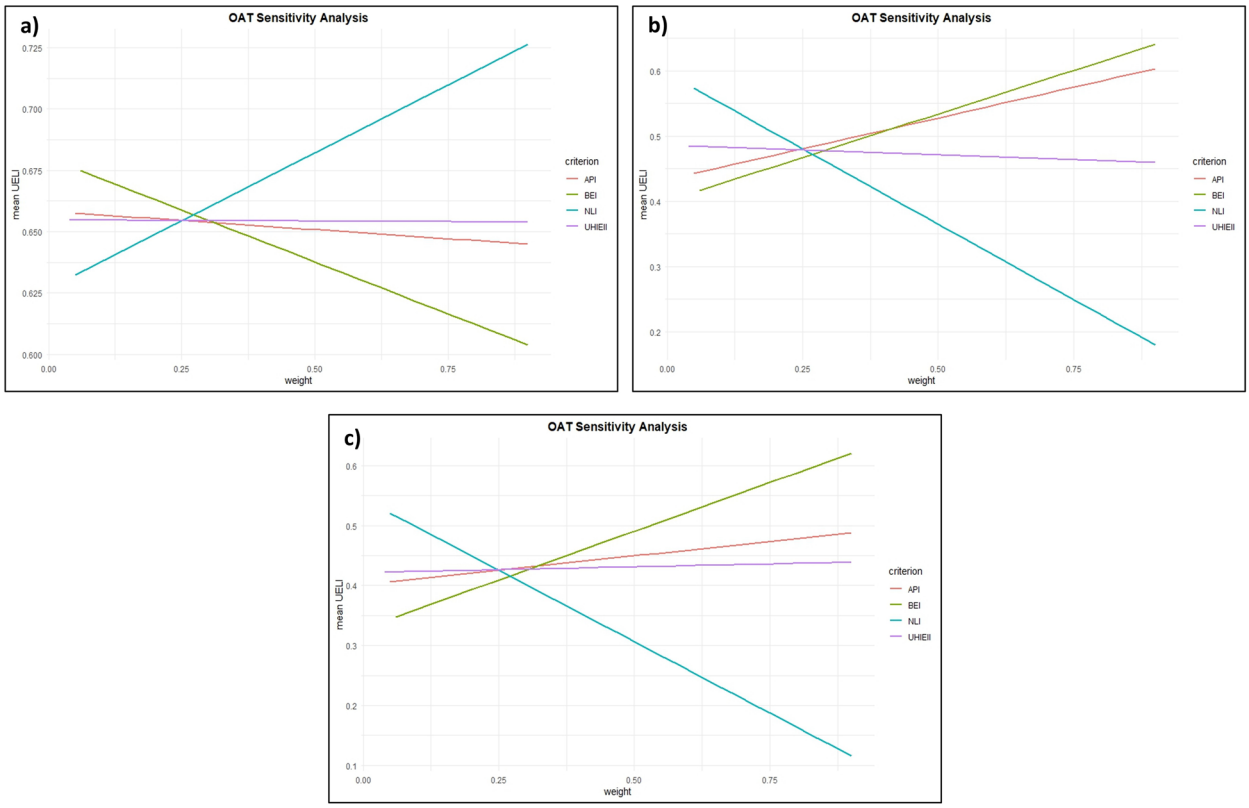


Fig. 10. OAT sensitivity plot analysis for UELI parameters (**a**) 2004, (**b**) 2014, and (**c**) 2024. These plots were generated using R software (version 4.5.2).

Validation and sensitivity analysis

The Global Sensitivity- PRCC and OAT plot used to identify how UELI has been influenced by multiple parameters, such as BEI, API, NLI, and UHIEII over 2004, 2014, and 2024 years (Fig. 10). The dual sensitivity approach tried to evaluate the stability and the reliability of UELI.

Figure 10, the OAT sensitivity plot analysis focused on individual parameter BEI, API, NLI, and UHIEII to identify the linearity and stability of UELI value in 2004, 2014, and 2024. In 2004, the BEI, UHIEII, and API parameter lines remain stable with a gentle upward slope, while the NLI and BEI parameters show the steepest slope, indicating that as the weight of API increases but the mean UELI value showed vise-versa of the indicators. In 2014, both NLI slope goes to downward as long as the weight increases and the mean value of UELI degrades. The API and BEI slope line significantly rises upward but UHIEII remain slightly steady. In 2024, the NLI slope remained declining meanwhile API and UHIEII goes to upwards with a close conducted value, and BEI remain steady. The BEI is employed as a positive driver; the API is identified as a negative friction in the entire analysis.

In this study, the PRCC analysis (Fig. 11) evaluates the correlation between different parameters of the UELI in a global perspective. Here, the PRCC was employed with 95% Confidence Intervals (CI) to identify the contribution of each driver. In 2004, the NLI and UHIEII were maintained as positive drivers with 0.939 and 0.269. On the other hand, the BEI and the API were determined as the negative drivers, which is -0.840 and -0.078. The NLI contributed as a positive dominant driver. But in 2014, the BEI shifted from a negative value to a positive value of 0.679. As well as NLI shifted to a negative value of -0.944 (Table 10). The API and UHIEII value shifted to 0.487 and -0.562. In 2024, the BEI was determined as the primary positive dominant driver with

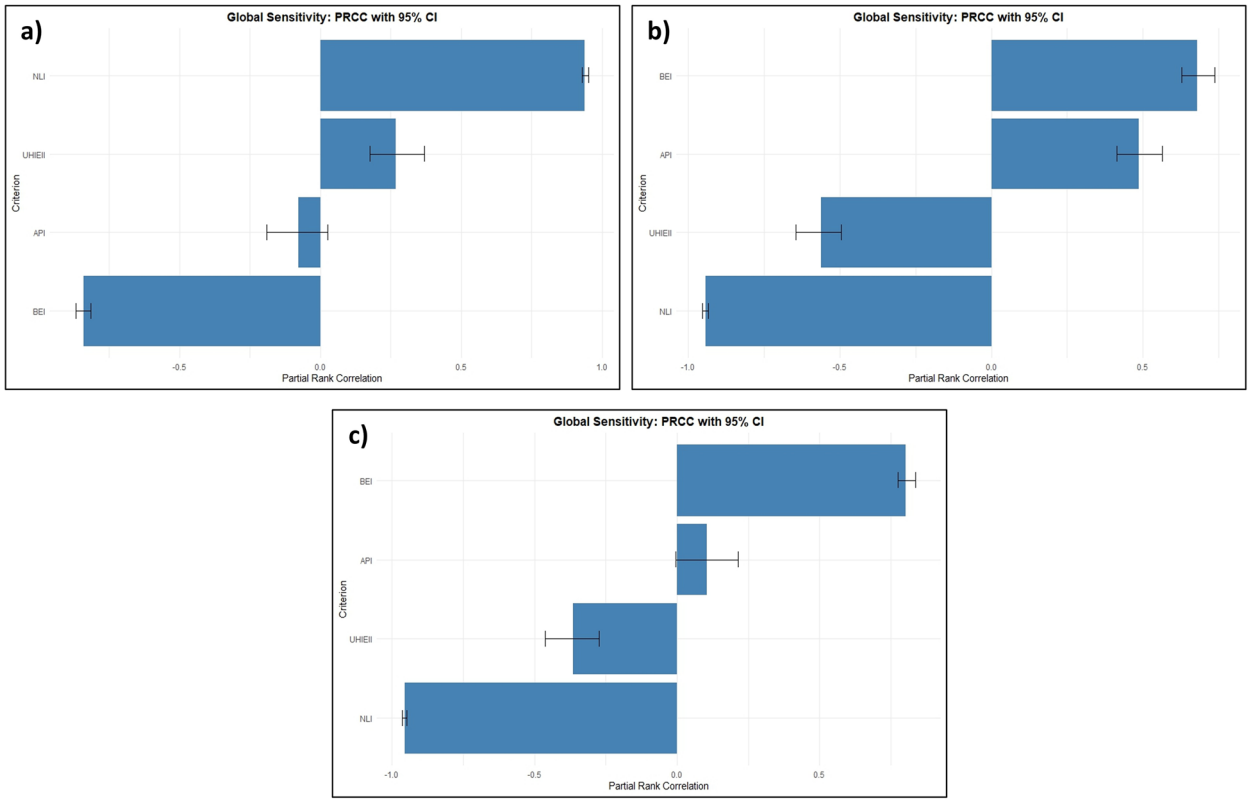


Fig. 11. PRCC plot analysis with CI for UELI parameter (**a**) 2004, (**b**) 2014, and (**c**) 2024. These plots were generated using R software (version 4.5.2).

	2004			2014			2024		
Criterion	PRCC estimate	Lower CI	Upper CI	PRCC estimate	Lower CI	Upper CI	PRCC estimate	Lower CI	Upper CI
BEI	−0.840	−0.869	−0.840	0.679	0.627	0.736	0.804	0.775	0.836
API	−0.078	−0.189	0.026	0.487	0.563	0.563	0.106	−0.005	0.214
NLI	0.939	0.929	0.951	−0.944	−0.954	−0.934	−0.954	−0.963	−0.947
UHIEII	0.269	0.176	0.369	−0.562	−0.645	−0.495	−0.364	−0.462	−0.273

Table 10. PRCC sensitivity analysis for the different parameters of UELI.

a value of 0.804. The API and NLI were consistently identified as a frictional parameter with a correlation value of 0.106 and -0.954, and the UHIEII remained as a negative correlation of -0.364 in PRCC.

Discussion

This study evaluates the urban livability conditions of Dhaka, a rapidly growing megacity. We employed several techniques: PCA, FAHP, spatial autocorrelation, scientific validation analysis, urban condition change detection analysis. We used PCA For statistical analysis to simplifying the high-dimensional data specially, the first principal component (PC1) was selected as to representing urban livability and environmental quality. We used the FAHP which is a multi-criteria decision-making (MCDM) technique using the triangular fuzzy numbers to calculate the weight distributions of the study area which tried to find out the livable condition of Dhaka & make a strong contribution in this context. Spatial autocorrelation identified the global and local spatial relationship of the existing urban environment indicators. Global Moran's I indicated that the urban dynamic indicators were statistically significant while LISA revealed that central, south-western part of study area addressed as high-high clustered zone through the spatial analysis. From the overall UELI change in Dhaka from 2004 to 2024, livable area addressed in vulnerable condition along with 30.81% as unlivable condition in 20 years. PRCC and OAT sensitivity analysis analyzed that the dynamicity of the parameters was the key responsible for the complex matrix and validate the intricate outcome of UELI.

In the study, seven deterministic components were evaluated, NDVI, Index- based IBI, Wetness, BEI, UHIEII, API, & NLI. The data of the image were evaluated after normalizing the image value from 0 to 1. One statistical test (PCA) was conducted to improve the image coordination by reducing an initial set of variables to create a smaller new set of orthogonal (uncorrelated)[90]. The assessment of FAHP analysis for crucial for

arranging the components in a hierarchical structure to identify the weight distribution of the factors[91]. In this study Fuzzy AHP was calculated for urban environmental sustainability with mathematical weights for BEI, API, UHIEII, and NLI that were determined to be 0.30, 0.25, 0.20, and 0.25, respectively. In another study, Fuzzy AHP was computed for the three components are social, economic, and environmental sustainability. Recent years different scholars have been using the various complex statistical technique and remote sensing aerial photograph to identify the evaluation of urban livability condition[92,93]. We employed the UELI and BEI to identify the urban condition with accuracy from 2004 to 2024 in our study. These statistical analyses investigate the study area, generating a closely outcomes to make a correlation and a variation in urban conditions in past 20 years. Compared to other studies, this analysis identifies convergent outcome, despite the utilize of parametric differences and challenging due to its limited use of parameter to developing the UELI map od Dhaka area[94]. However, through GEE platform is significant for spatial and temporal analysis with extract remote sensing data to calculate the UELI[17].

We employed the UELI and BEI to identify the urban condition with accuracy from 2004 to 2024 in our study. These statistical analyses investigate the study area, generating a closely outcomes to make a correlation and a variation in urban conditions in past 20 years. A trend of spatial homogenization that threatens sustainable urban livability is depicted in the LISA cluster maps from 2004 to 2024. The High-High (HH) clustering examined as a consolidated distribution in both 2004 and 2014. But by 2024, an advanced UHI impact is indicated by the dispersed of High-High (HH) clusters into a stiff central, south-western core. This density in Dhaka is caused by the quick transformation of floodplains into impermeable surfaces, which exacerbates the UHI effect and reduces inhabitants' comfort in the microclimate[95]. A critical disparity in Ecosystem Service Values (ESV) is seen in this east–west difference. Although the enduring Low-Low (LL) clusters in the east serve as regional cooling sinks, fair access to natural cooling is hindered by their complete isolation from the H–H zones. In order to interrupt the continuity of heat-dense clusters, safe-to-fail techniques are needed to achieve urban resilience. Specifically, green–blue infrastructure must be interwoven throughout the urban center[96].

A significant spatiotemporal shift has been identified in transition of UELI analyzing the environmental indicators from 2004 to 2024. This distribution characterized by heavily persistence of less livable condition accounting for 17.33%, and 17.66% from 2004 to 2014 between 2014 to 2024 respectively. A critical finding showed a dramatical transition towards unlivable condition 6.89% to 15.22% and finally expanded to 24.69% from 2004 to 2014 and 2024 respectively. This study identified the persistent with recent study showed that rapid and unplanned urbanization and land use transition frequently intensified the ecological loss, exposure to heat stress, Landscape fragmentation. Strong restoration and planning intervention are required to be implemented on urban dominant ecological degradation and urban environmental evolution[97]. UELI validation was supported by utilizing PRCC and OAT sensitivity analysis with 95% CI. Global PRCC measures the independent effect of each livable indicator while OAT measured the temporally transition of the indicators and its response.

Ultimately, the outcome of the study clearly shows that the core part of the study area faces the impact of rapid and unplanned urbanization regarding the year variation. The urban livable condition should be considered alongside the limitation overviewed in limitation section, despite of this, the study holds a significant consequence for governments and policymakers developing a sustainable plan for mitigating the rapid urbanization. This research might allow the people of Dhaka city to be cautious of possible actions to avoid the adverse effects of rapid urbanization and reduce the ecological risk of the urban environment.

Policy implementations

This research findings provide a guidance for the both urban planners and the policymakers to work on balancing between rapid urbanization and the ecological risk. This paper recommended that the Local Government focuses low livable condition area to enhance the environmental quality as it degraded gradually. This study's findings can be use in to upgrade the quality of livable condition, economic growth and environmental health, the LGD and Dhaka City Corporation significant contribution can mitigate the heat stress, vegetation loss, environmental pollution, unplanned urban develop and expansion. The findings emphasize on the comprehensive and integrated planning framework and implemented projects such as Dhaka Structure Plan 2016–2035, the Detailed Area Plan (DAP) to create a more efficient and resilient urban form. Department of Environment (DoE) of Ministry of Environment, Forest and Climate Change (MoEFCC), leads national policy formulation, Regional Development Planning (RDP) act as a leading contributor from planning to feasible condition through improve land management, structural plan zoning, multi-modal transportation with connectivity, access to green space, improve micro-climate which are key to urban livability.

Non-governmental organizations such as BRAC, ESDO, and BEDS etc. prioritize the fundamental layer of policy and implementation by promoting community based environmental quality analysis, inclusive public environments awareness, open space protection etc. to support the sustainable urban planning and mitigate the urban ecological risk.

Recommendations for future studies

Significant research challenges of this study identified and must be investigated to improve the existing UELI condition and its change modelling. Further research may address these critical gaps and strengthening future research directions in different scale:

1. Avoiding the temporal coarseness and mismatch with study intervals limits, Census data from 2001, 2011 and 2022 was not used in the study because of the dynamic nature of Dhaka's demographic data.
2. Air quality assessment needs to produce more accurate exposure through comprehensive pollutant-level (NO_2, SO_2, O_3, CO) analysis integrating both ground monitoring, and Remote sensing observations.

3. Harmonize remote sensing data with dynamic LULC change of Dhaka city micro-scale drive an insightful environmental heterogeneity with linking the socio-economic layers.
4. Conduct robust spatiotemporal ML frameworks with uncertainty estimation to integrate accurate, policy-relevant exposure and vulnerability maps and rigorous model diagnostics.

Conclusion

This study aimed to conduct urban livability mapping in the capital city of Bangladesh, Dhaka, known as a densely populated city with rapid unplanned urban growth. The results indicate a progressive decline in urban livability conditions over the study period. The proportion of highly livable area in Dhaka city was 17.78%, 15.89%, & 12.93% in 2004, 2014, and 2024, respectively, declining with the ecological parameter. Around 10.73% livable area degraded to unlivable while only 0.37% and 2. 51% area improved to very livable and livable from unlivable condition from in past 20 years. High-High clusters have been demonstrated by Global Moran's I and LISA maps, indicating increased heat stress and a significant decrease in the UELI in the western part of the city. The UELI change detection indicated the rapidly transition of urban livable condition across these years. The key factors influencing livability were changes in Built-up areas. The substantially decreased region could have been identified by the local government and focus on enhancing the livability condition of the city by taking the necessary steps to increase the quality of the environment, living condition, avoid disasters, and promote economic growth. The adverse effect of the urban heat island still has to be mitigated in the core part of the city. The research findings provide policymakers with comprehensive knowledge that supports planning and decision-making for sustainable urban planning and mitigating the urban ecological risk.

Data availability

Data used in this research are available from the corresponding author upon reasonable request. Ethical approval for animal research: This study did not involve any experiments on animals.

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Author contributions

Khandoker Mariatul Islam: Data curation; Formal Analysis; Methodology; Validation; Visualization; Roles/Writing—original draft; and Writing—review & editing; Md. Mahmudul Hasan: Conceptualization; Supervision; Data curation; Formal analysis; Investigation; Resources, Methodology; Software; Validation; Visualization; Roles/Writing—original draft; and Writing—review & editing; Afroja Khanam Sonia, Mahzabin Akhter & Barbara Sneha Gomes: Visualization; Roles/Writing—original draft; Writing—review & editing; Most. Mitu Akter: Software; Validation; Visualization; Roles/Writing—original draft; and Writing—review & editing; N. M. Refat Nasher: Writing—review & editing.

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Declarations

Competing interests
The authors state that there are no financial or personal conflicts of interest that could have influenced this work.

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